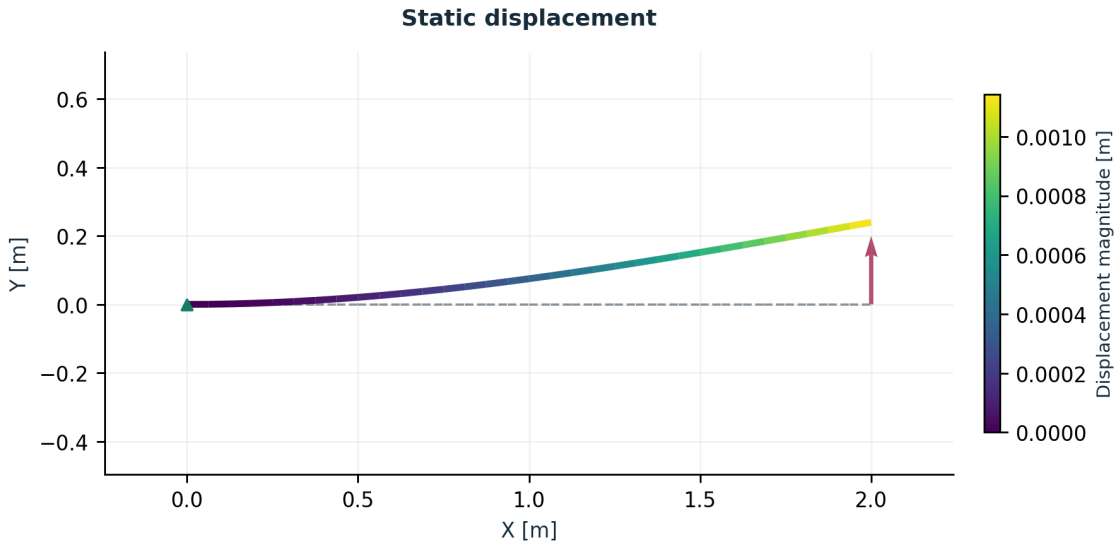
Slender cantilever: static bending and natural frequencies

ACA01 | SOL 101, 103 | Independent analytical equation verification

A clamped uniform beam tests exact static bending and the first four Euler-Bernoulli normal modes. Consistent translational mass and a single bending plane give an independently specified one-dimensional eigenproblem.

Geometry / mesh	Length 2 m; section 0.04 m by 0.01 m. Meshes of 4, 8, 16 and 32 CBAR elements.
Material / section	E = 70 GPa, nu = 0.3, density 2700 kg/m3; A = 4e-4 m2, I = 3.333333333e-9 m4. Shear deformation disabled.
Boundary conditions	One bending plane: every node ux=uz=rx=ry=0; root uy=rz=0.
Applied load	Static: +Y tip force 0.1 N. Modal: no load; consistent translational mass.



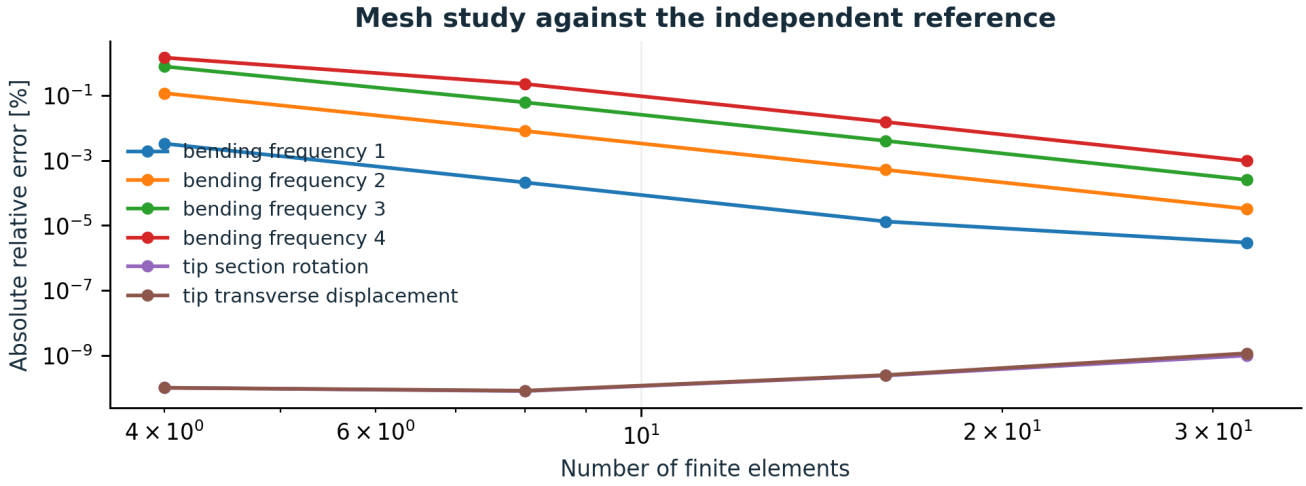
Shape amplified x210; grey = original.

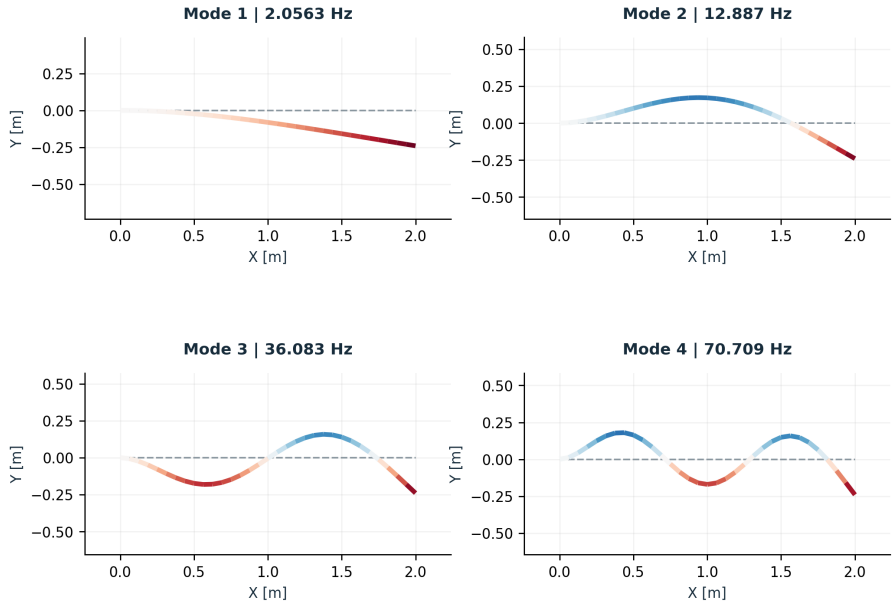
Actual solver output. Deformation is amplified for visibility; mode vectors have arbitrary sign and amplitude.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
tip transverse displacement [m]	0.0011429	0.0011429	1.155e-09%	PASS
tip section rotation [rad]	0.00085714	0.00085714	9.8153e-10%	PASS
bending frequency 1 [Hz]	2.0563	2.0563	2.939e-06%	PASS
bending frequency 2 [Hz]	12.887	12.887	3.2104e-05%	PASS
bending frequency 3 [Hz]	36.083	36.083	0.00025119%	PASS
bending frequency 4 [Hz]	70.709	70.708	0.00096223%	PASS

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

Independent reference equations

w(x)=F*x^2*(3L-x)/(6EI)

theta(x)=F*x*(2L-x)/(2EI)

cos(beta)*cosh(beta)=-1

f_j=beta_j^2/(2*pi*L^2)*sqrt(EI/(rho*A))

All four finest frequencies agree within 0.000963%; static deflection and rotation agree to roundoff.

The fourth-frequency error falls from 1.4522% with 4 elements to 0.0009623% with 32 elements.

Scope: Euler-Bernoulli equation verification with small displacement and one bending plane; no experimental calibration or general three-dimensional beam certification.

Reference and reproduction

University of Alberta: Euler Bernoulli Beam
https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/plane-beam-approximations/euler-bernoulli-beam/

Slater and Timbo: Engineering Vibration Toolbox, continuous systems
https://vibrationtoolbox.github.io/vibration_toolbox/reference/continuous.html

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997cce40 | Full precision values: results.json / metrics.csv